

SAKSHI SAH

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SUMMARY

Computer Science undergraduate (CGPA: 9.06) building production AI/ML systems—from high-throughput semantic search pipelines to scalable hybrid retrieval engines. Experienced in orchestrating multi-stage multimodal architectures, vector databases, and fault-tolerant backend infrastructures with hackathon-winning system design proficiency.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript (ES6+), SQL, Java, C
- **AI/ML & NLP:** PyTorch, TensorFlow, Scikit-learn, Sentence Transformers, Cross-Encoders, Multimodal LLM Orchestration, Pandas, NumPy, Plotly
- **Databases & Vector Stores:** ChromaDB, Qdrant, MySQL, Redis (Caching)
- **Web Frameworks & Tools:** FastAPI, Streamlit, React.js, Flask, REST APIs, Git, GitHub Actions, Docker, Linux, Render Cloud, Vercel, AWS
- **Core Computer Science:** Data Structures & Algorithms (DSA), System Design (LLD/HLD), Object-Oriented Programming (OOP), Operating Systems, DBMS, Computer Networks

EDUCATION

- **Bachelor of Technology in Computer Science & Engineering** | KIIT University | Expected 2027 | **CGPA: 9.06/10**
- **CBSE (Class XII)** | VJ International School | 2022 | **85.4%**
- **ICSE (Class X)** | St. Joseph's School | 2020 | **89.0%**

TECHNICAL PROJECTS

CircularAI – Intelligent Reverse Logistics Platform | *Amazon Hackathon*

- **Stack:** FastAPI, Python, Gemini API (Multimodal LLM), Asyncio, Pydantic v2
- Orchestrated an asynchronous FastAPI backend utilizing multimodal LLM architectures to automate resale condition grading (Good/Fair/Poor) on high-volume product streams.
- Designed a deterministic multi-stage operational pipeline handling image quality assertion, damage detection vector evaluation, dynamic routing logic, automated inventory listing, and telemetry calculation for carbon offsets.
- Conducted black-box regression testing across 10 separate commercial product categories, validating cross-category schema consistency and deterministic grading profiles under continuous parallel loads.
- Links: GitHub: <https://github.com/SAKSHI1243/second-life-commerce>
- Live Demo: <https://second-life-commerce-lac.vercel.app/>

PatentLens AI – Hybrid Prior-Art Retrieval Platform

- **Stack:** ChromaDB, Rank-BM25, Cross-Encoder Transformers, Streamlit Cloud, Pysqlite3
- Engineered a robust two-stage hybrid retrieval search platform executing parallel BM25 lexical token matching and dense semantic vector embeddings over 1,700+ discrete patent text fragments.
- Built a high-performance second-stage re-ranking engine using Cross-Encoder models (`ms-marco-MiniLM-L-6-v2`) to compute mathematical alignment probabilities, shifting inference execution entirely to CPU to reduce memory footprints by 95%.
- Implemented strict connection overrides (`pysqlite3-binary`) to bypass native cloud Linux container constraints, ensuring seamless server-side database serialization and stateless operational uptime.
- Optimized UI multi-tab rendering cycles via custom Streamlit `session_state` persistence arrays, ensuring search contexts and similarity matrices survive execution loop changes without flushing cache memory.
- Links: ks:[GitHub] <https://github.com/SAKSHI1243/patent-lens-copilot>
- Live Demo: <https://patent-lens-copilot-dz2lmhk66htmww8jenely54.streamlit.app/>

PhishGuard – ML-Based Threat Intel & Phishing Detection Tool

- **Stack:** FastAPI, Streamlit, Scikit-learn, Joblib, NetworkX, Render Cloud
- Developed a decoupled microservices architecture isolating an interactive Streamlit UI configuration from a stateful, high-throughput FastAPI machine learning prediction backend.
- Implemented automated extraction engines parsing 88 distinct statistical, network (SSL validation, WHOIS age telemetry), and structural features, achieving a 96.24% classification accuracy over a reference pool of 11,430 targets.
- Configured asynchronous fallback handlers and exception blocks preventing system-wide thread blocking during slow DNS lookups or socket-level timeouts.

- Links: [GitHub](https://github.com/SAKSHI1243/phishguard-system)- <https://github.com/SAKSHI1243/phishguard-system>
- | [Live Demo](https://phishguard-dashboard-smjr.onrender.com) -<https://phishguard-dashboard-smjr.onrender.com>

HireSense AI – Resume-to-Job Matching System

- **Stack:** Python, Sentence Transformers, Scikit-learn, PyPDF2, Plotly, Streamlit
- Programmed an text-mining semantic engine converting unstructured PDF files into normalized high-dimensional vectors via pre-trained Transformer layers to compute precise cosine similarity intersections.
- Refactored a defensive PDF parsing middleware layer utilizing structured text boundary heuristics, recovering incomplete or malformed content frames across 90% of corrupted legacy document buffers.
- Automated unit validation matrices via local testing profiles to verify data consistency across iterative pipeline version increments.
- **Links:** Links: [GitHub](<https://github.com/SAKSHI1243/HireSense-AI>) |
 - [Live Demo](<https://hiresense-ai-y4hxzmkz7kl7phxs53crkg.streamlit.app/>)

ACHIEVEMENTS & CERTIFICATIONS

- **Data Structures & Algorithms:** Solved **350+ data structure and algorithmic challenges** across LeetCode and GeeksforGeeks; proficient in Dynamic Programming, Graph Theory, and Tree/Matrix traversals.
- **Smart India Hackathon (SIH):** College-level **Runner-Up (2nd out of 502 teams)**. Engineered the high-concurrency system architecture and rate-limiting patterns for a crowdsourced civic alerting system.
- **Certifications:** Oracle Cloud Infrastructure Certified Foundations (Oracle University), IBM AI Fundamentals, Tata Generative AI Professional Certificate.